ENTERPRISE ENGINEERING BLUEPRINT

ContentOS SMM Suite v2.0

Software Requirement Specification (SRS)

End-to-End Social Media Automation: PDF Knowledge & Visual Ingestion, Multi-Platform Variant Generation, Day-by-Day Drip Scheduler, Indian Festival Integration, and Dual Notifications (In-App + Registered Gmail).

AUTHOR / ORGANIZATION

ContentOS / Hindustan Social

TARGET ENVIRONMENT

Node.js · NestJS · React 19 · BullMQ

DOCUMENT STATUS

Approved for Implementation

1. Executive Summary & Vision (मुख्य उद्देश्य)

The goal of this initiative is to eliminate manual social media management completely by creating a fully autonomous, intelligent content engine. The system transforms unstructured assets (bulk manual inputs or rich multi-page PDFs with text and embedded images) into ready-to-post, platform-optimized creatives that publish on a daily automated schedule.

> **Core Requirement Highlight:** Dual-mode input ingestion, non-broken visual extraction, platform-specific copy (LinkedIn, Instagram, Facebook, Telegram, YouTube), automated daily drip feed, autonomous festival greeting injection, and multi-channel notifications (In-app SMM panel badge + direct Gmail delivery to the user's signup email).

2. Current State vs. Target State (Gap Analysis)

Capability	Current Status in Workspace	Target Requirements	Action Required
PDF Image Previews	Broken icon: `alt="PDF image from page X"`	Crisp, instant thumbnail rendering in UI grid	Fix signed download token & auth check on local storage route
Input Modes	Manual single post or basic PDF chunking	Dual Input: Bulk Manual Editor + Multi-PDF Document Engine	Provide unified bulk upload and auto-variant generation
Variant Generation	Text copy generation only	Complete bundle: 1080×1080 Poster + Caption + Description + Hashtags	Connect canvas poster generator with extracted PDF visuals
Daily Posting	Manual per-post scheduling	Automatic Drip Feed: sequentially assigns slots across upcoming days	Implement automated slot allocator into BullMQ queue
Festival Auto-Post	Calendar engine exists in `server/calendar/`	Auto-injects branded festival posters into daily schedule	Priority auto-scheduling without exceeding daily capacity limits
Email Notifications	Auth OTP emails only	Instant Gmail delivery on post publish or failure	Integrate publishing event hooks with `email_outbox` worker

3. Module 1: Knowledge & Visual Ingestion Pipeline

Mode A: Bulk Manual Creation

Allows content creators to paste or upload CSVs with multiple drafts (Title, Body, Target Date, Media URLs). The system validates limits and stores all drafts directly in the queue.

Mode B: Multi-PDF Document Extraction (Text + Visuals)

- **Text Chunker & AI Citations:** Analyzes research papers, brochures, product specs, and generates verifiable claims.
- **Embedded Image Extractor (Screenshot Issue Resolution):**
  In the current system, images extracted from PDFs show broken image icons because the browser's `<img src="...">` does not transmit custom workspace headers to `/api/v1/storage/local/download`. The fix will attach secure, HMAC-signed query tokens to preview URLs, allowing image previews to load instantaneously without authentication dropouts.
- **1-Click Image Reuse:** Extracted high-res images can be designated as the primary creative asset for generated social variants.

4. Module 2: AI Multi-Platform Variant & Creative Engine

Once content or PDF insights are extracted, the system auto-generates platform-specific variations placed into the **"Generated Variants"** repository:

Platform	Tone & Architecture	Char Limit	Asset Specification
LinkedIn	Executive thought leadership, structured spacing, 3-5 tags	3,000	Branded text card or extracted PDF visual
Instagram	High hook rate, engaging line breaks, 15-20 targeted tags	2,200	Square 1080×1080 JPEG (Canvas rendered)
Facebook	Conversational community tone, link previews allowed	63,206	Landscape / Square Graphic + CTA link
Telegram	Markdown typography, instant broadcast callouts	4,096	Photo banner with formatted caption
X (Twitter)	Punchy summary, 2-3 trending hashtags	280	Single high-impact visual attachment

5. Module 3 & 4: Daily Drip Scheduling & Festival Automation

The platform eliminates the need to schedule posts manually one by one. Once variants are generated, hitting **"Auto-Schedule Drip"** initiates:

1. **Slot Calculation:** Checks workspace predefined posting slots (e.g., 10:00 AM, 03:00 PM, 08:00 PM).
2. **Sequential Distribution:** Assigns pending post bundles sequentially to Day 1, Day 2, Day 3, and so on.
3. **Indian & Global Festival Detection:** The system consults `festival.data.ts`. If an upcoming date is a recognized festival (e.g. Diwali, Eid, Independence Day, Holi):
   - An autonomous festival poster is rendered using `@napi-rs/canvas` with festival theme colors and workspace branding.
   - The greeting post is injected into that day's morning slot.
   - Regular content is smoothly shifted to the afternoon slot to respect the workspace daily limit without any collision.

6. Module 5: Dual-Channel Notifications (SMM Panel + Gmail)

Whenever an automated post executes (either successfully published or flagged with an error), notifications are dispatched via two synchronized channels:

1. **`Publishing Event:`** `BullMQ publishing worker completes provider execution on Instagram, LinkedIn, FB, or Telegram.`
2. **`Channel A - SMM Panel:`** `Real-time row added to notifications table. Badge counter ticks up; user can click to view direct live post link.`
3. **`Channel B - Gmail Delivery:`** `Worker queries the registered account email (signup email) and queues a branded email via Nodemailer SMTP.`

Email Alert Specifications

- **Success Template:** ✅ `[Published] Your scheduled post is live on {Platform}` with direct link, execution timestamp, and post preview.
- **Failure Alert:** ⚠️ `[Action Needed] Post publishing failed on {Platform}` containing the exact error reason (e.g. token expired, media size exceeded) and 1-click reconnect button.

7. Implementation Roadmap & Verification

Phase	Deliverable	Verification Metric
Phase 1	Fix PDF image previews & storage download authentication	PDF page thumbnails render cleanly without broken icons
Phase 2	AI multi-platform variants generator UI integration	"Generated Variants" stores all 5 platforms with 1-click approval
Phase 3	Automated Day-by-Day drip scheduler + Festival injection	Posts queue across consecutive days; festival greeting injected
Phase 4	Dual-channel publishing notifications (SMM bell + Gmail)	Post execution sends instant email to registered user's Gmail